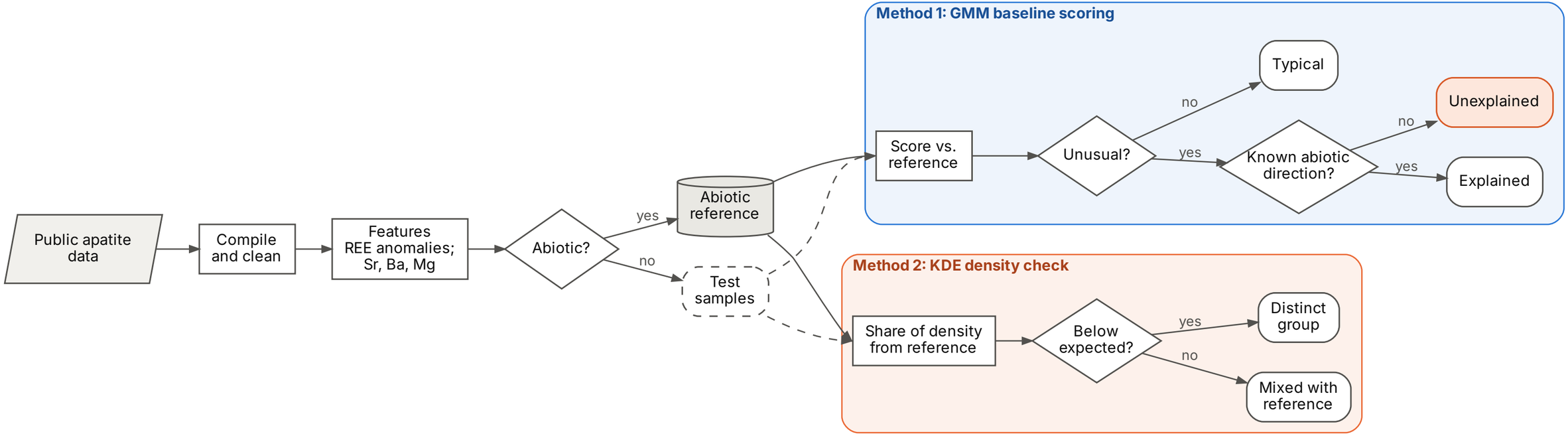
Public apatite data
Compile and clean
Features REE anomalies; Sr, Ba, Mg
Abiotic?
yes
no
Abiotic reference
Test samples
Method 1: GMM baseline scoring
Score vs. reference
Unusual?
no
Typical
yes
Known abiotic direction?
no
Unexplained
yes
Explained
Method 2: KDE density check
Share of density from reference
Below expected?
yes
Distinct group
no
Mixed with reference